

Intern Project

8-point Radix-2 SDF DIF FFT Accelerator

Omar Hussien & Bassel Mahmoud

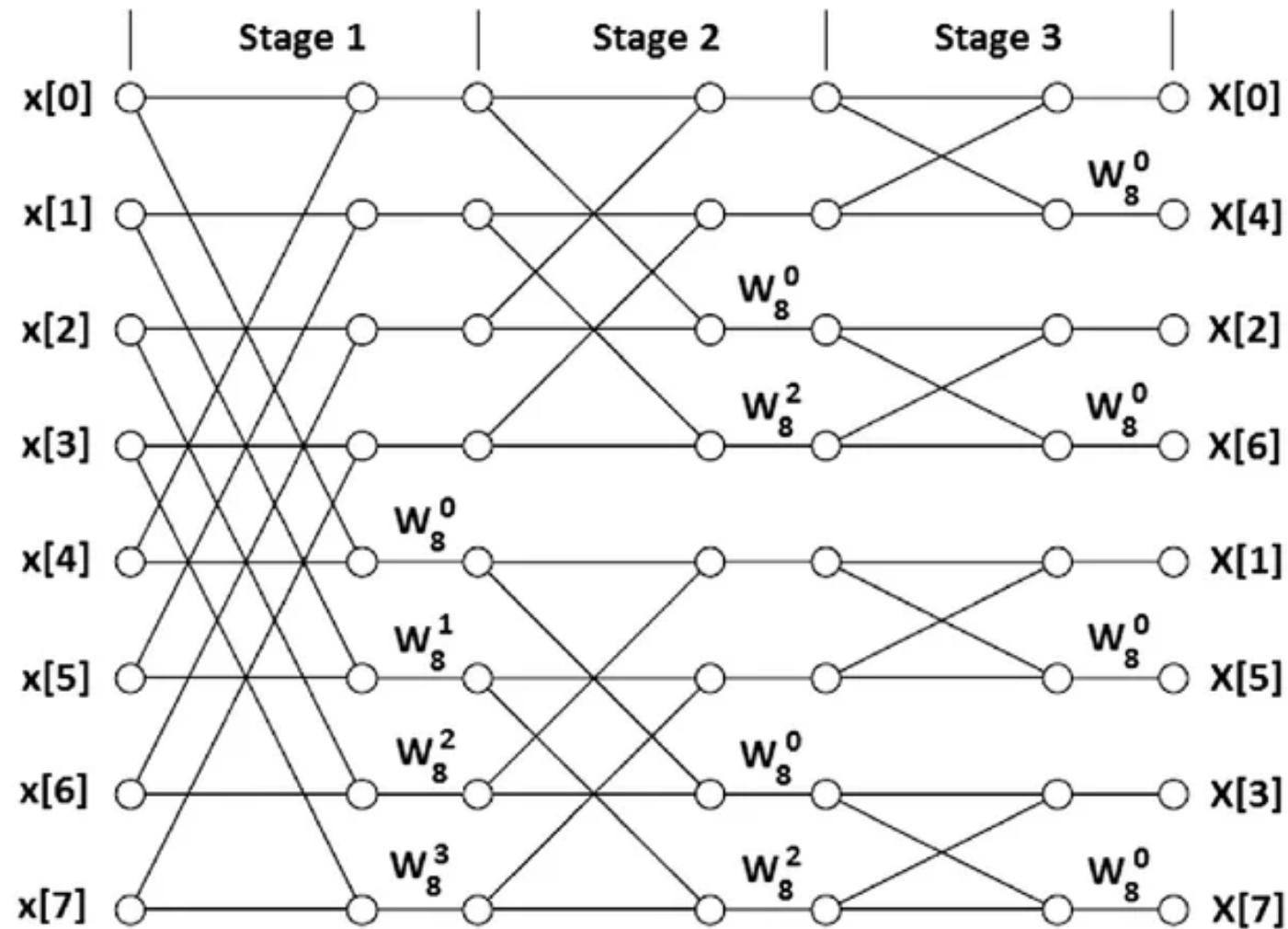
System/Digital Design Interns

- Project Overview.
- System Modeling , Fixed-Point Model & Signal Sizing.
- RTL Design , Architecture & Latency.
- RTL Verification — UVM & Coverage

Project Overview

Parameter	Choice
N	8 points
Algorithm	Radix-2
Decimation	DIF
Architecture	SDF Pipeline

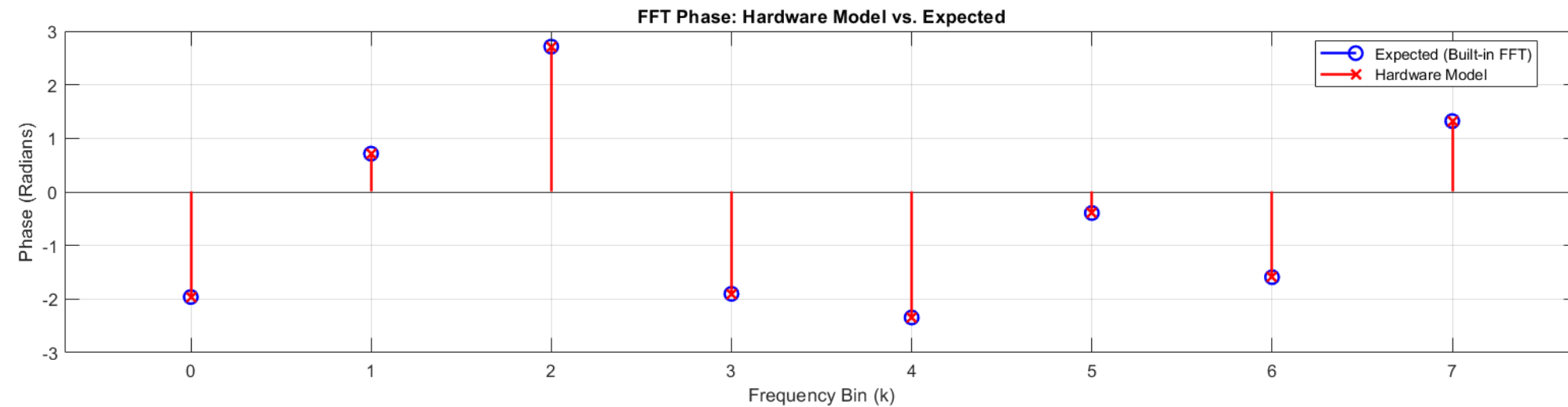
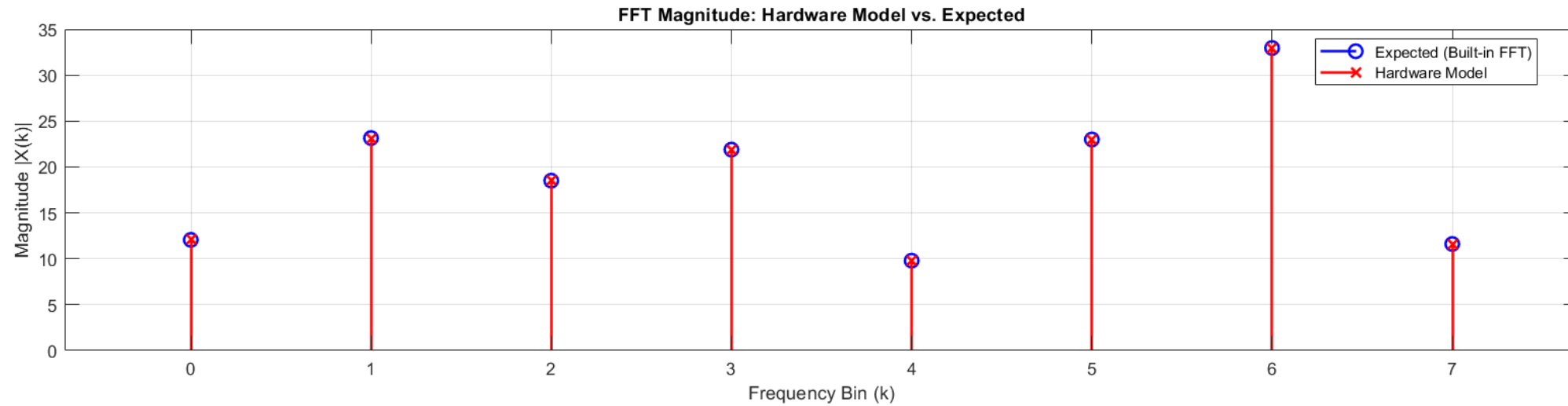
8-Point DIF Flow chart



System Modeling , Fixed-Point Analysis & Signal Sizing.

```
1 function [ADD1, SUB1, MULT1, ADD2, SUB2, MULT2, ADD3, SUB3, MULT3] = fft_sdf_core_try(DIN_STREAM, ROM_OUT_1, ROM_OUT_2,T)
2     %#codegen
3
4     ADD1 = cast(DIN_STREAM(1:4) + DIN_STREAM(5:8) , 'like' ,T.ADD1 );           % 4 element vector
5     SUB1 = cast(DIN_STREAM(1:4) - DIN_STREAM(5:8) , 'like' ,T.SUB1 );           % 4 element vector
6     MULT1 = cast(SUB1 .* ROM_OUT_1 , 'like' ,T.MULT1);           % 4 element vector
7
8     S2_top = [ADD1(1:2); MULT1(1:2)];
9     S2_bottom = [ADD1(3:4); MULT1(3:4)];
10
11     ADD2 = cast(S2_top + S2_bottom , 'like' , T.ADD2 );           % 4 element vector
12     SUB2 = cast(S2_top - S2_bottom , 'like' , T.SUB2 );           % 4 element vector
13     MULT2 = cast(SUB2 .* ROM_OUT_2 , 'like' , T.MULT2);           % 4 element vector
14
15     S3_top = [ADD2(1);ADD2(3); MULT2(1);MULT2(3)];
16     S3_bottom = [ADD2(2);ADD2(4); MULT2(2);MULT2(4)];
17
18     ADD3 = cast(S3_top + S3_bottom , 'like' , T.ADD3 );           % 4 element vector
19     SUB3 = cast(S3_top - S3_bottom , 'like' , T.SUB3 );           % 4 element vector
20     MULT3 = cast(SUB3 , 'like' , T.MULT3);           % 4 element vector
21 end
```

FFT Model Verification



Fixed Point analysis with Matlab

Instrumentation Report Viewer - D:\Omar_Hussein\College\ADI intern\Sessions\Project\Matlab files\instrumentation\fft_sdf_core_mex\html\report.mldatx

REPORT

Back Forward Go To Find Edit In MATLAB EDIT

NAVIGATE

MATLAB Source

Function List Call Tree

fft_sdf_core.m
fx fft_sdf_core

```
Function: fft_sdf_core
1 function [ADD1, SUB1, MULT1, ADD2, SUB2, MULT2, ADD3, SUB3, MULT3] = fft_sdf_core(DIN_STREAM, ROM_OUT_1, ROM_OUT_2,T)
2     %#codegen
3
4     ADD1 = cast(DIN_STREAM(1:4) + DIN_STREAM(5:8) , 'like' ,T.ADD1 );    % 4 element vector
5     SUB1 = cast(DIN_STREAM(1:4) - DIN_STREAM(5:8) , 'like' ,T.SUB1 );    % 4 element vector
6     MULT1 = cast(SUB1 .* ROM_OUT_1 , 'like' ,T.MULT1);    % 4 element vector
7
8     S2_top = [ADD1(1:2); MULT1(1:2)];
9     S2_bottom = [ADD1(3:4); MULT1(3:4)];
10
11     ADD2 = cast(S2_top + S2_bottom , 'like' , T.ADD2 );    % 4 element vector
12     SUB2 = cast(S2_top - S2_bottom , 'like' , T.SUB2 );    % 4 element vector
13     MULT2 = cast(SUB2 .* ROM_OUT_2 , 'like' , T.MULT2);    % 4 element vector
14
15     S3_top = [ADD2(1);ADD2(3); MULT2(1);MULT2(3)];
16     S3_bottom = [ADD2(2);ADD2(4); MULT2(2);MULT2(4)];
17
18     ADD3 = cast(S3_top + S3_bottom , 'like' , T.ADD3 );    % 4 element vector
19     SUB3 = cast(S3_top - S3_bottom , 'like' , T.SUB3 );    % 4 element vector
20     MULT3 = cast(SUB3 , 'like' , T.MULT3);    % 4 element vector
21 end
```

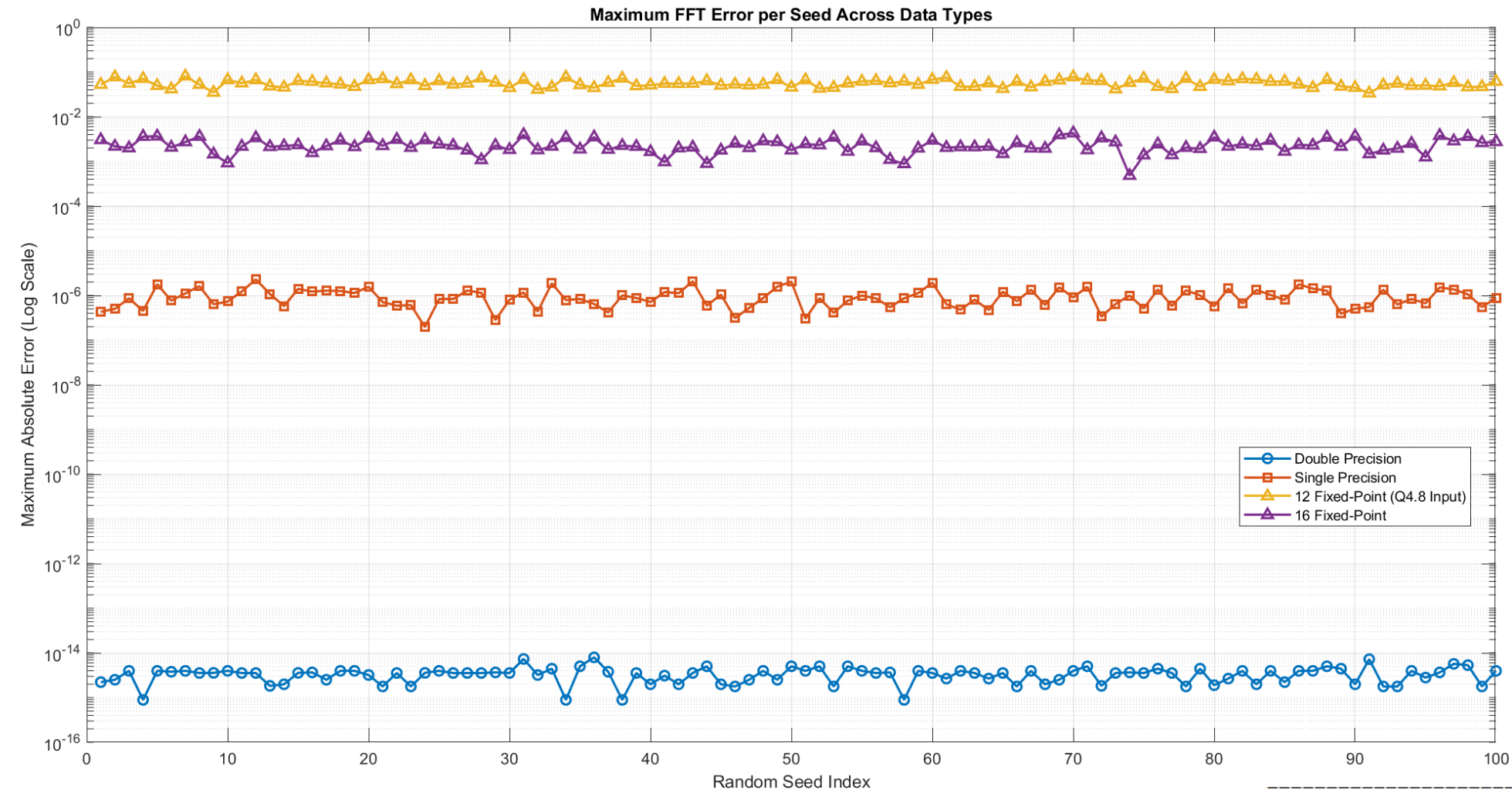
All Messages (0)

Name	Type	Size	Class	Proposed Signedness	Proposed WL	Proposed FL	Always Whole Number	Sim Min	Sim Max
ADD1	Output	4 × 1	complex double	Signed	32	27	No	-15.21875	15.9375
ADD2	Output	4 × 1	complex double	Signed	32	26	No	-24.221169391503263	26.524790702087564
ADD3	Output	4 × 1	complex double	Signed	32	25	No	-39.092915601876896	38.78125
MULT1	Output	4 × 1	complex double	Signed	32	26	No	-17.429077301902794	15.976193837433557
MULT2	Output	4 × 1	complex double	Signed	32	26	No	-22.7734375	25.83984375
MULT3	Output	4 × 1	complex double	Signed	32	25	No	-41.42578125	43.30859375
SUB1	Output	4 × 1	complex double	Signed	32	27	No	-15.10546875	15.328125

Signal Sizing Results

Signal Name	Sim Min	Sim Max	Word Length	Fraction Length	Integer Bits	Proposed WL	New Fraction Bits
DIN_STREAM	-8	7.99609375	32	28	4	12	8
ROM_OUT_1	-1	1	32	30	2	12	10
ROM_OUT_2	-1	1	32	30	2	12	10
Signal Name	Sim Min	Sim Max	Word Length	Fraction Length	Integer Bits	Proposed WL	New Fraction Bits
ADD1	-15.21875	15.9375	32	27	6	12	7
MULT1	-17.4290773	15.97619384	32	26	6	12	6
SUB1	-15.1054688	15.328125	32	27	5	12	7
Signal Name	Sim Min	Sim Max	Word Length	Fraction Length	Integer Bits	Proposed WL	New Fraction Bits
ADD2	-24.2211694	26.5247907	32	26	6	12	6
MULT2	-22.7734375	25.83984375	32	26	6	12	6
SUB2	-22.7734375	25.83984375	32	26	6	12	6
Signal Name	Sim Min	Sim Max	Word Length	Fraction Length	Integer Bits	Proposed WL	New Fraction Bits
ADD3	-39.0929156	38.78125	32	25	7	12	5
MULT3	-41.4257813	43.30859375	32	25	7	12	5
SUB3	-41.4257813	43.30859375	32	25	7	12	5

Error Plots & SQNR across different signal widths



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AVERAGE SQNR REPORT

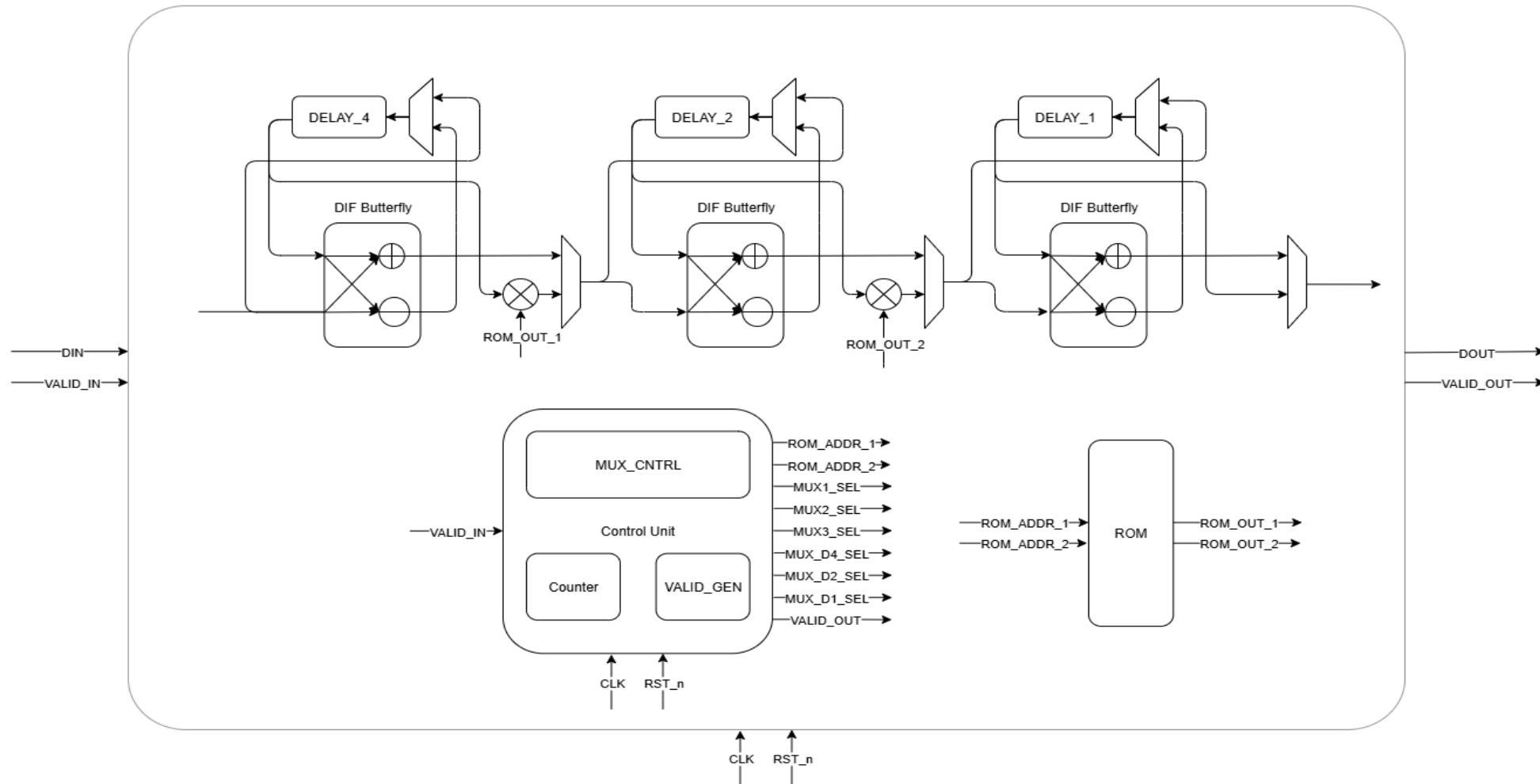
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Average SQNR (12-bit Datapath): 55.20 dB
Average SQNR (16-bit Datapath): 84.31 dB

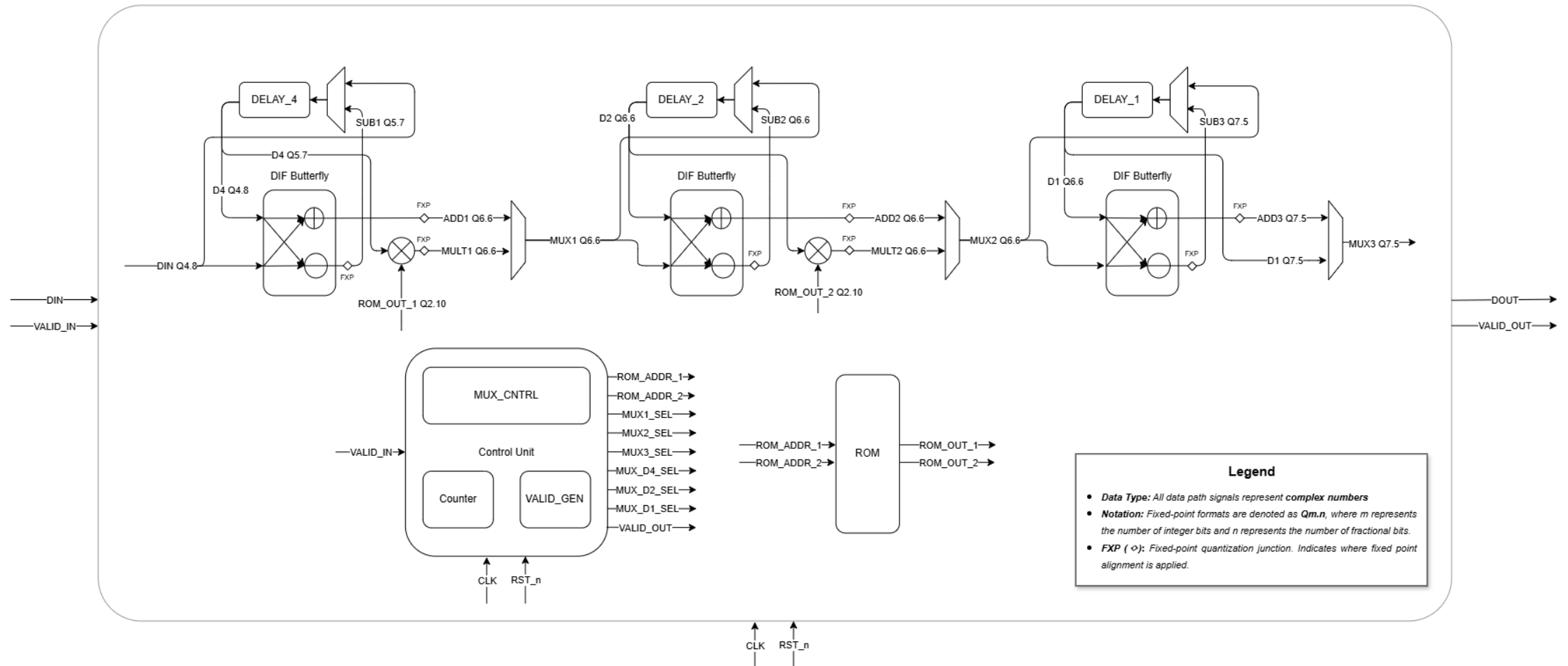
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RTL Design , Architecture & Latency

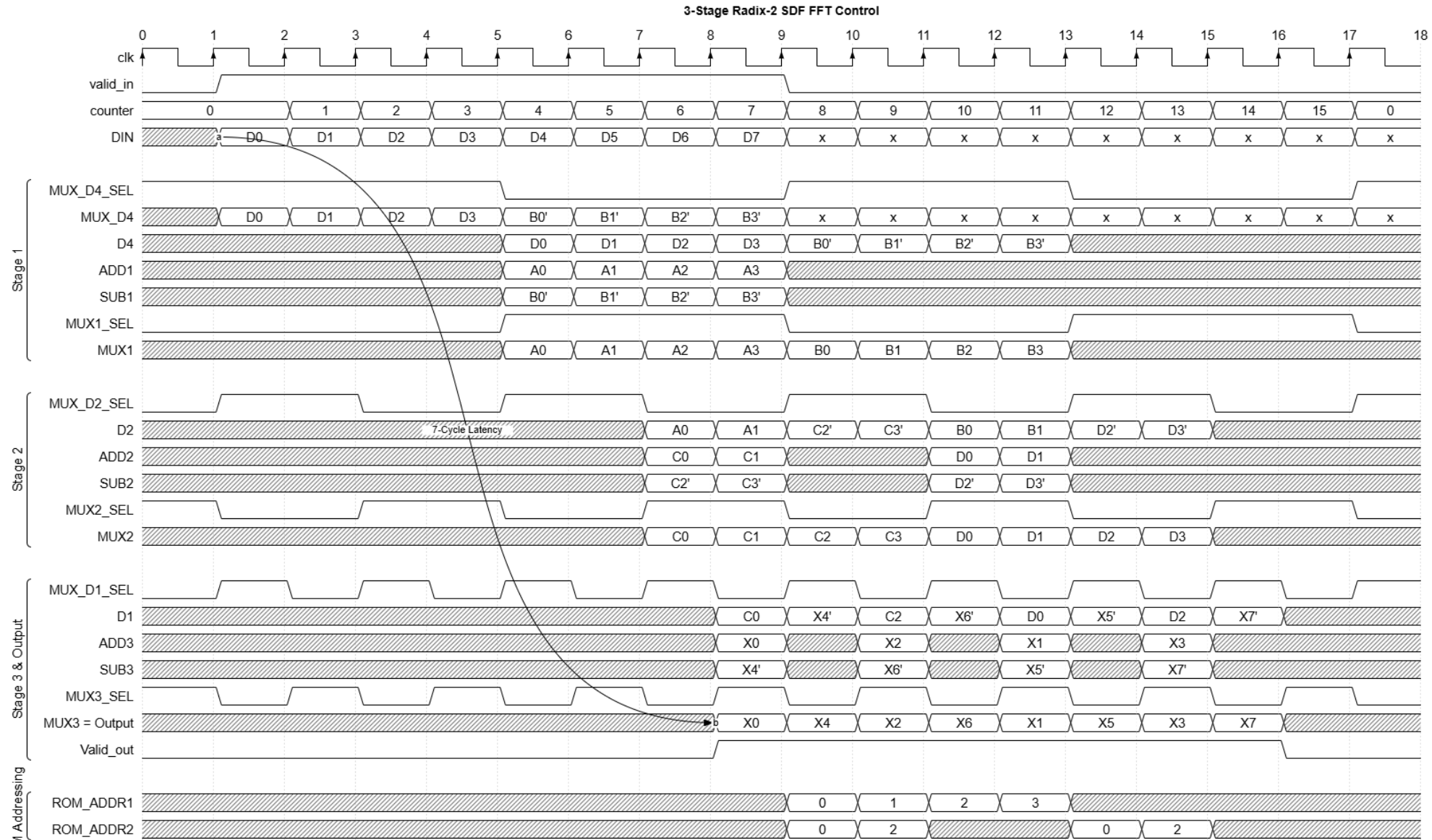
Architecture of a 8-Point Radix-2 SDF FFT Accelerator



Architecture of a 8-Point Radix-2 SDF FFT Accelerator

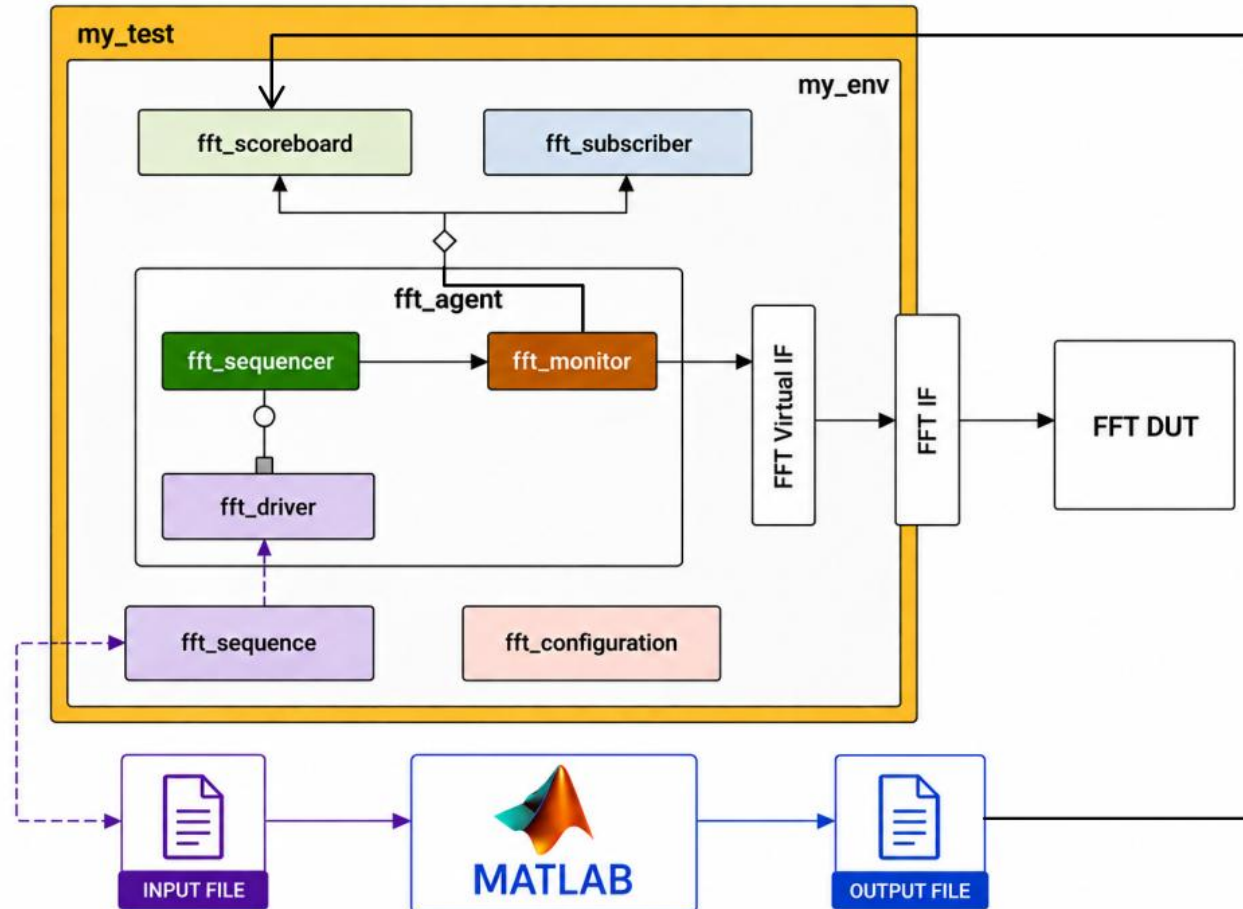


Pipeline & Latency Measurement



RTL Verification Using UVM

- Golden reference: MATLAB Model output used as the scoreboard reference

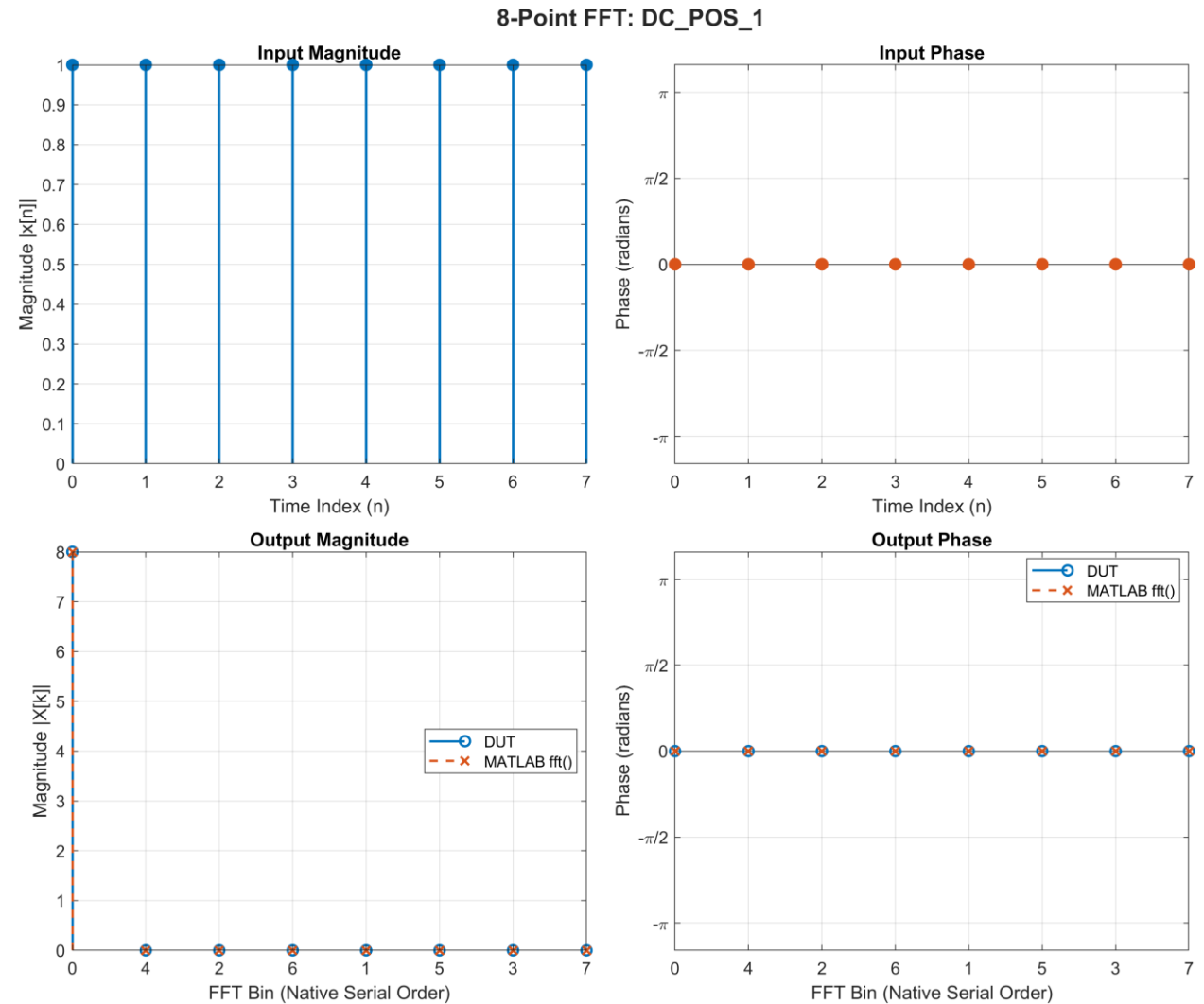


- Code Coverage : **100%**
- Functional Coverage : **100%**

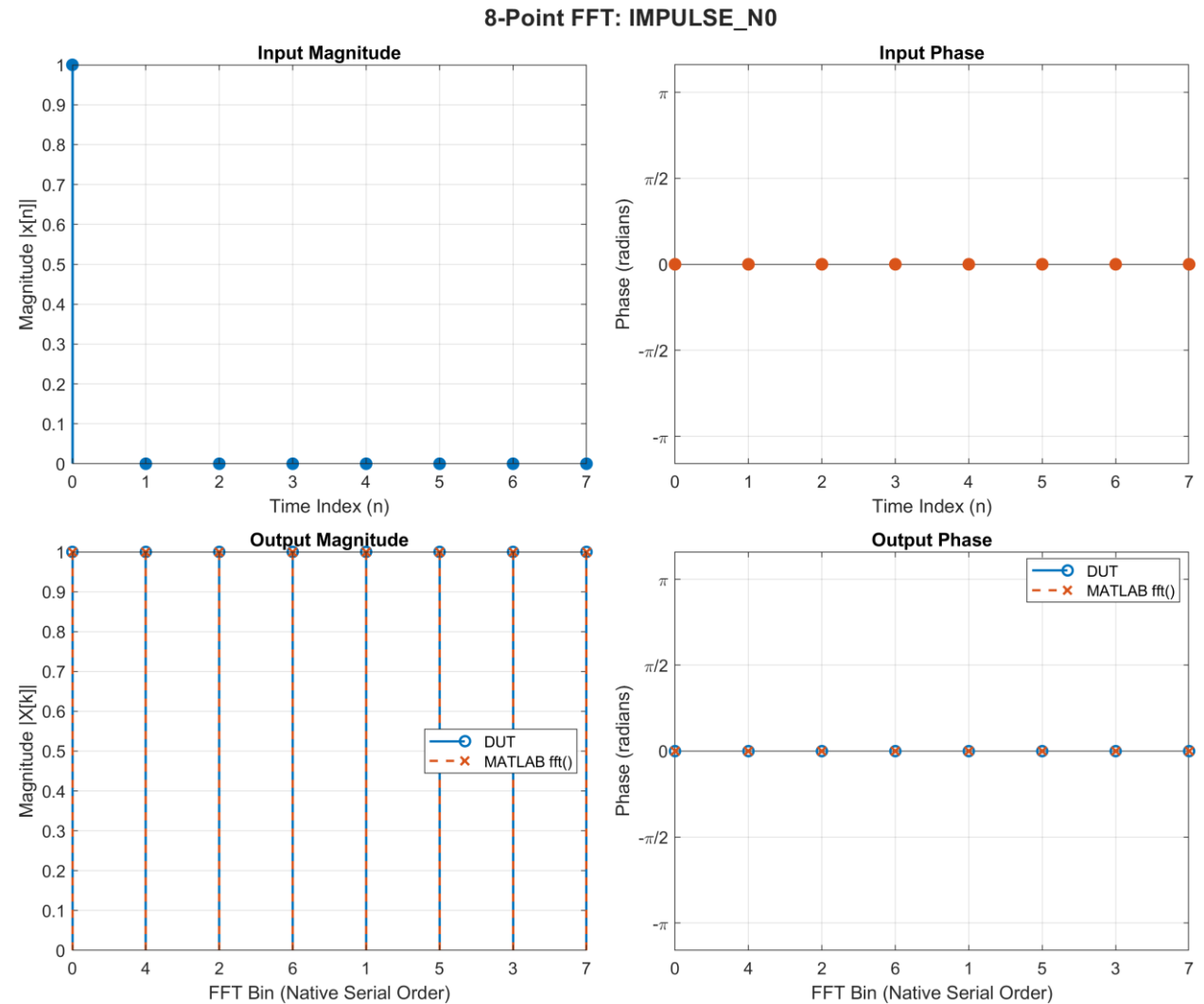
```
TOTAL COVERGROUP COVERAGE: 100.00%  COVERGROUP TYPES: 1
```

```
Total Coverage By Instance (filtered view): 100.00%
```

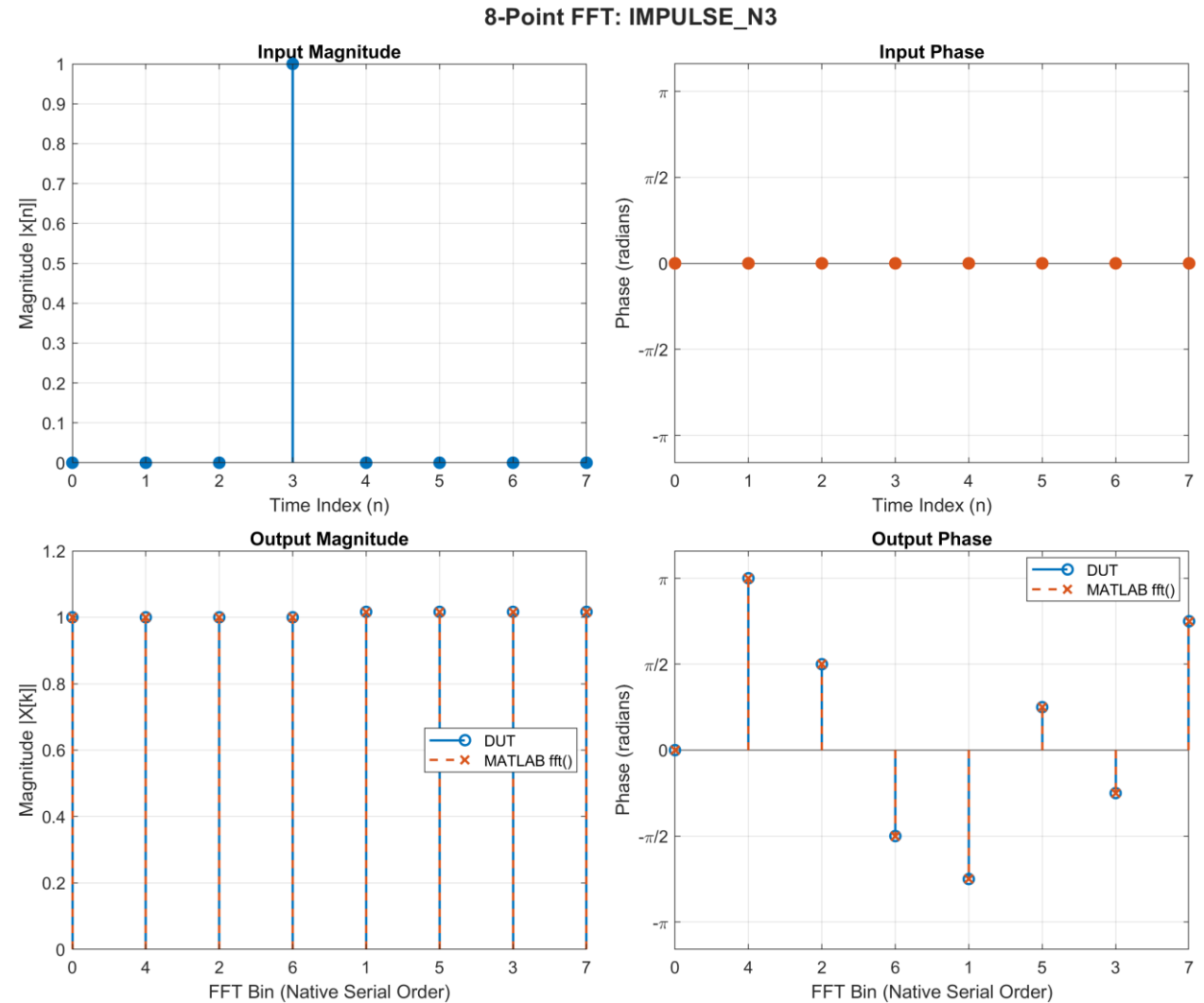
- DC Test:



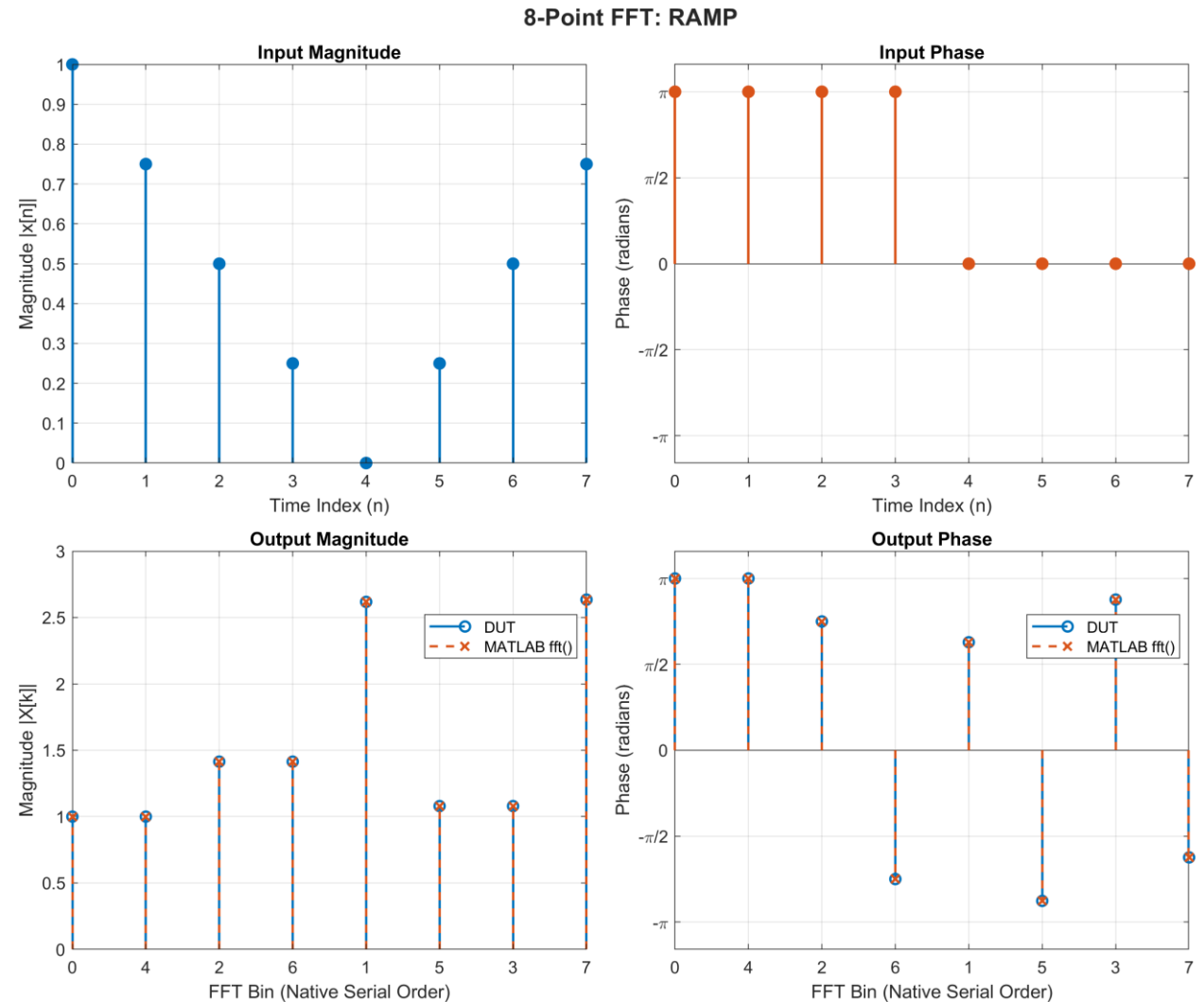
- Impulse Test:



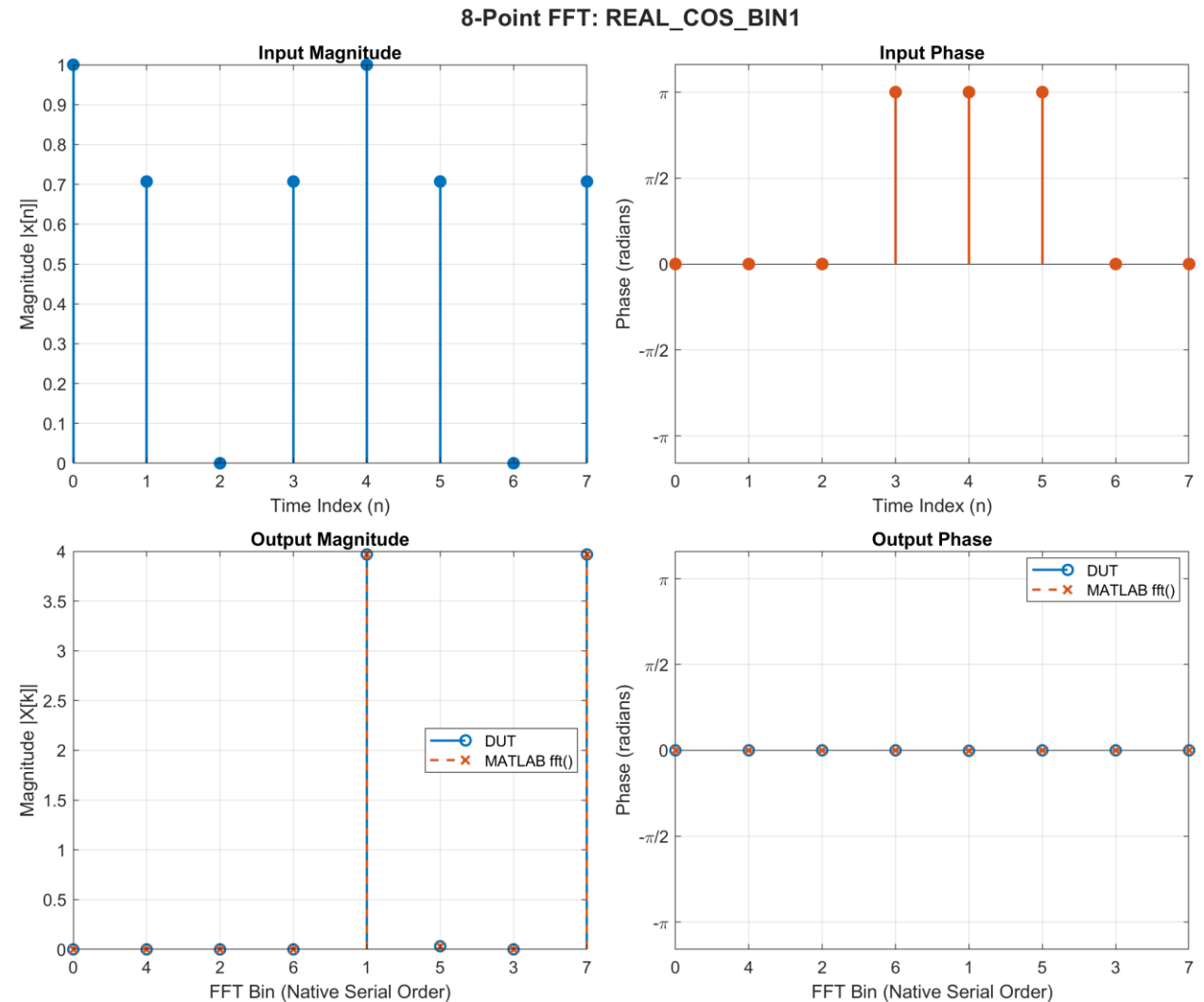
- Shifted Impulse Test:



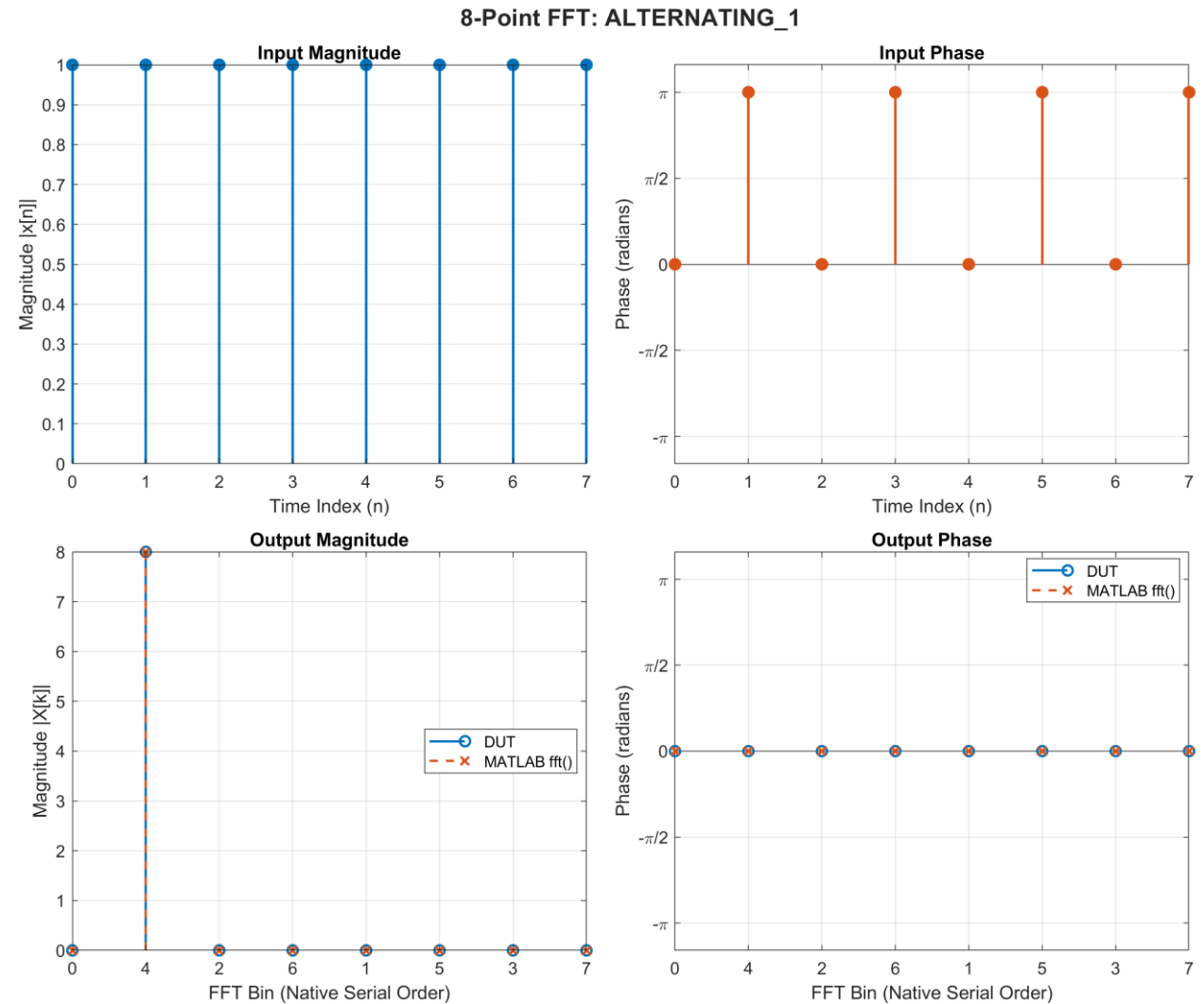
- Ramp Test:



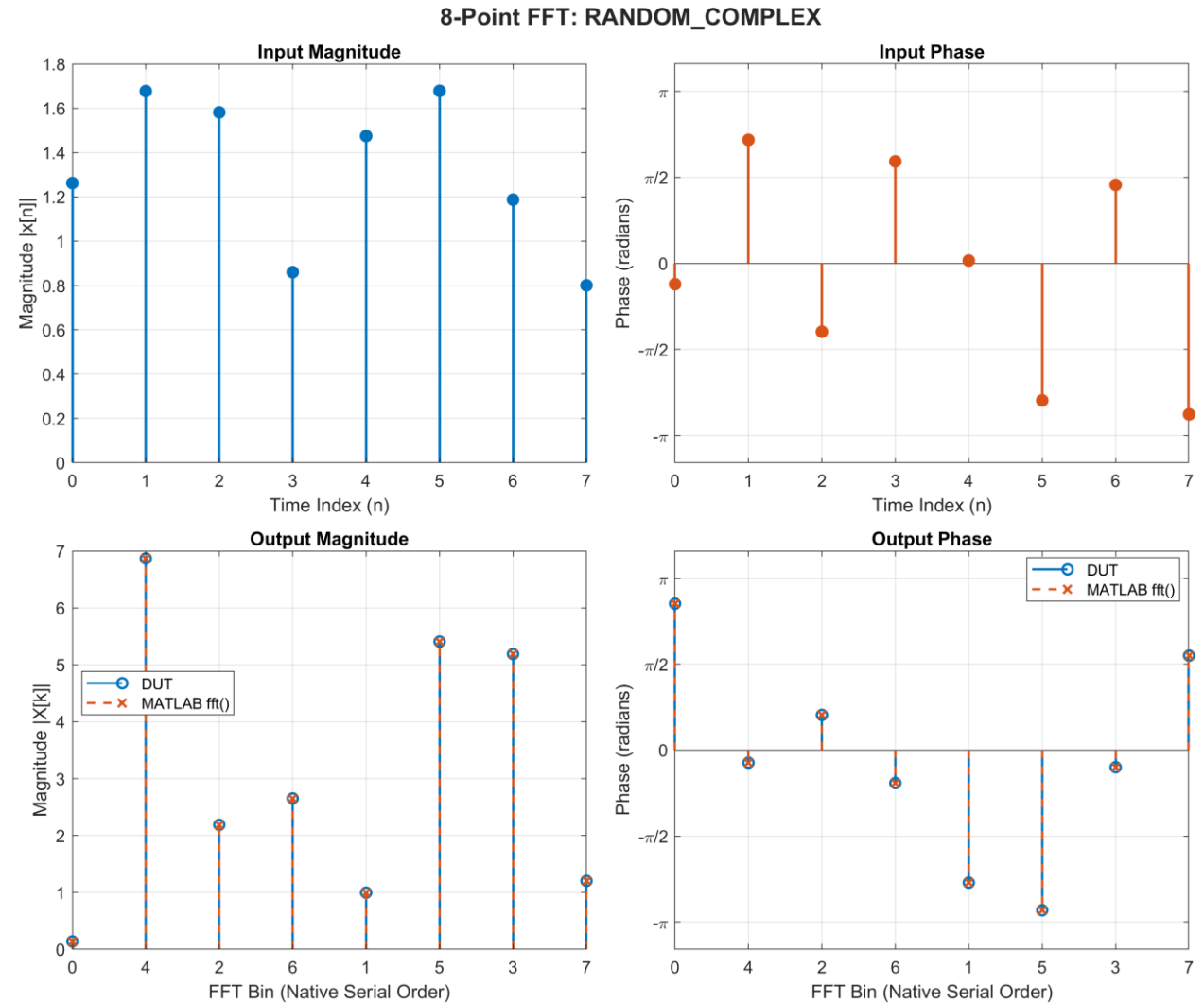
- Cosine Test:



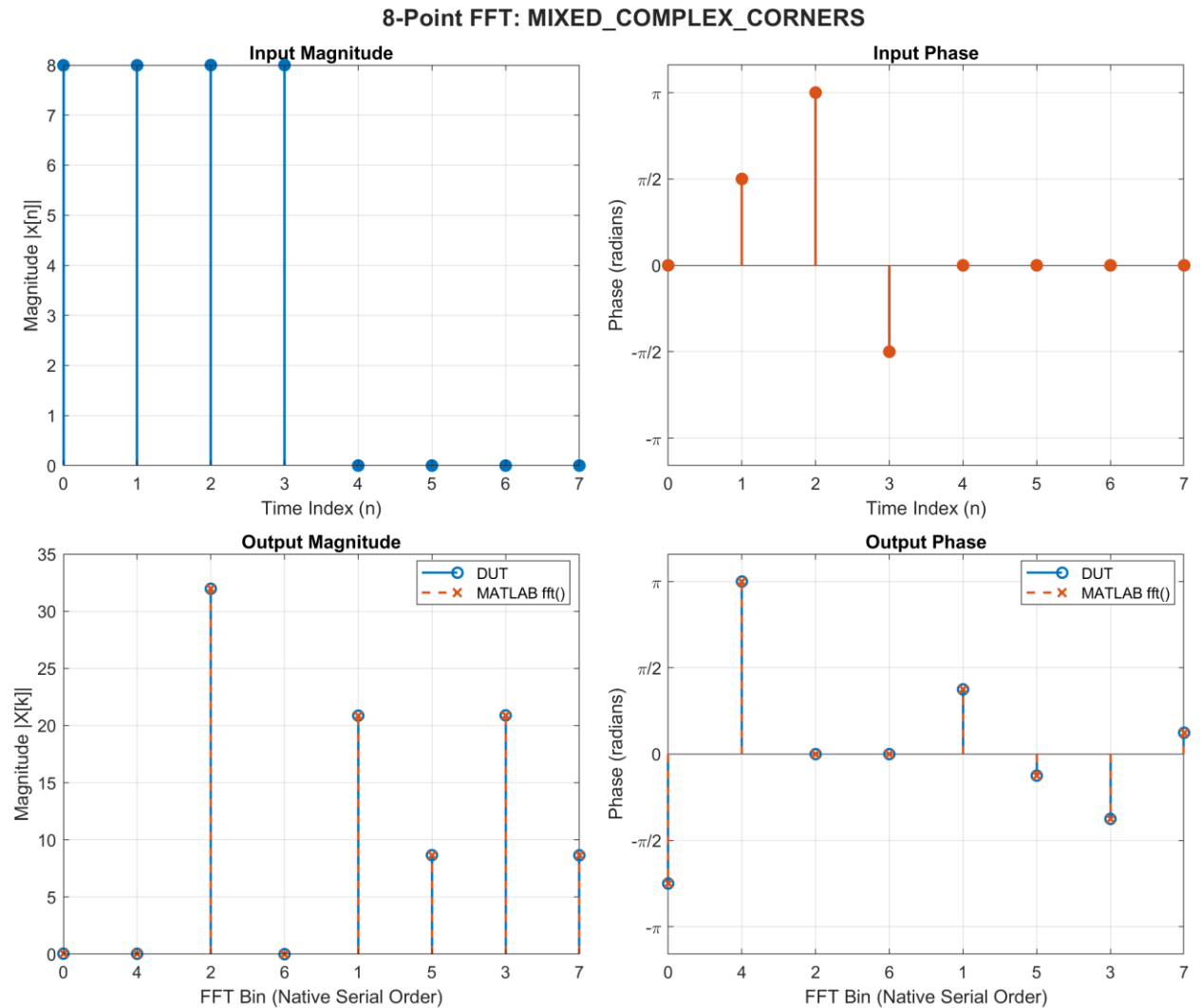
- Alternating Test:



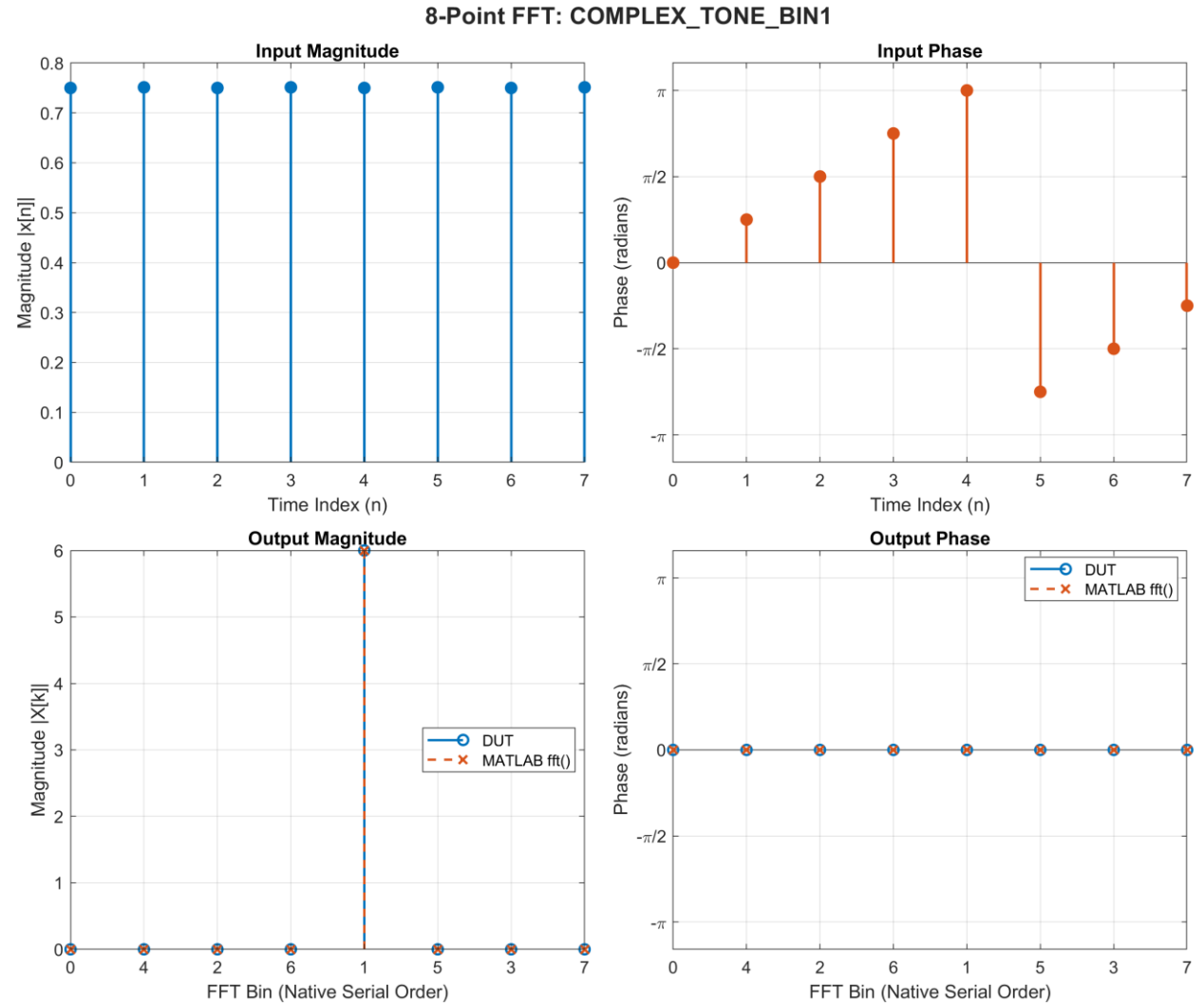
- Random Complex Test:



- Random Complex Test:



- Random Complex Test:



Summary

Phase	Result
System Modeling	Avg SQNR : 55.2 dB – 12 bit
RTL Design	Latency = 7 cycles
RTL Verification	Coverage : 100 %

Thank You!